**m.parthu (192312331)**

# Experiment No:1: Simulation of FDMA in MATLAB Date

## Program (Matlab Code)

```
clc; clear; close all;

fs=1000; t=0:1/fs:1;

TBW=600; Users=3;

CBW=TBW/Users;

fc=[100 300 500];  m=sin(2*pi*10*t);  s1=m.*cos(2*pi*fc(1)*t);

s2=m.*cos(2*pi*fc(2)*t);  s3=m.*cos(2*pi*fc(3)*t);  subplot(2,1,1)

plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on  title('FDMA Channel Allocation'),xlabel('Time

(s)'),ylabel('Amplitude') subplot(2,1,2)  bar(1:Users,CBW*ones(1,Users)),grid on

xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per

User')
```

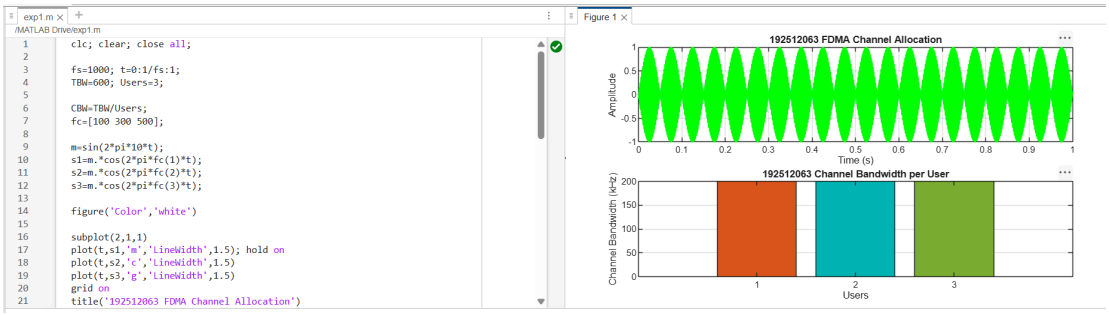



FIG.1.

## Objective 2:

```
clc; clear; close all;

BW = 200; % Channel Bandwidth (kHz)

Users = 3;
```

M = 2; % BPSK Modulation

DataRate = BW*log2(M); % Data Rate

(kbps)  Rate = DataRate*ones(1,Users);

disp(' User Data Rate (kbps)')

disp([(1:Users)' Rate'])  figure

bar(1:Users,Rate)  xlabel('Users')

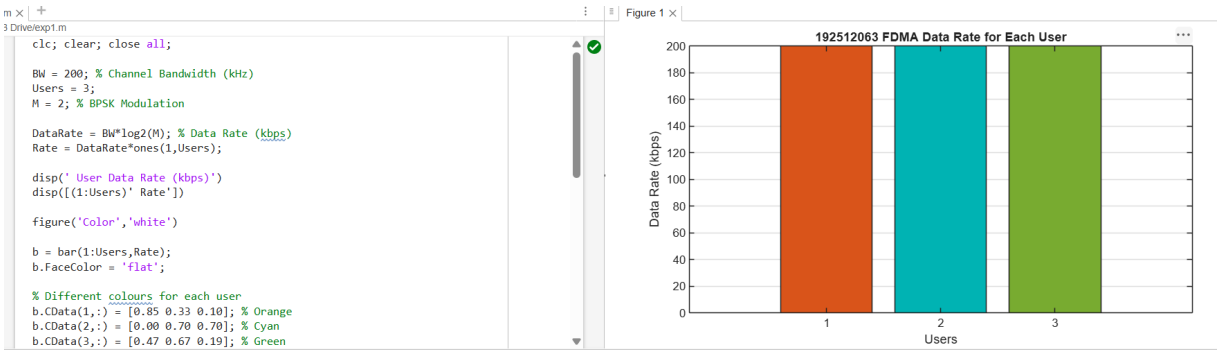
ylabel('Data Rate (kbps)')

title('FDMA Data Rate for Each User')

 grid on

FIG.2

.




clc; clear; close all;

Ps = 1; % Signal Power (W)

Pn = [0.1 0.05 0.01]; % Noise Power (W)

Users = 1:3;

SNR = 10*log10(Ps./Pn); % SNR in dB

disp('User Noise Power(W) SNR(dB)')

disp([Users' Pn' SNR'])  figure

bar(Users,SNR)  xlabel('Users')

ylabel('SNR (dB)') title('Signal-to-Noise

Ratio of FDMA Users') grid on

FIG.3.

```
clc; clear; close all;

Ps = 1; % Signal Power (W)
Pn = [0.1 0.05 0.01]; % Noise Power (W)
Users = 1:3;

SNR = 10*log10(Ps./Pn); % SNR in dB

disp('User Noise Power(W) SNR(dB)')
disp([Users' Pn' SNR'])

figure('Color','white')

b = bar(Users,SNR);
b.FaceColor = 'flat';

% Different colours for each user
b.CData(1,:) = [0.85 0.33 0.10]; % Orange
b.CData(2,:) = [0.00 0.70 0.70]; % Cyan
b.CData(3,:) = [0.47 0.67 0.19]; % Green
```

